Technical Project Report

Cross-Subject Generalization in EEG-Based Motor Imagery Classification

A Comparative Study of Subject-Dependent Learning, Leave-One-Subject-Out Cross-Validation, and EEGNet

BCI Competition IV-2a

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github.com/alitkbbl/Hybrid-EEG-Classifier-EEGNet

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Abstract

Brain–Computer Interfaces (BCIs) based on motor imagery (MI) EEG provide a non-invasive way of translating brain activity into control commands. A major challenge, however, is that the statistical structure of EEG signals can vary substantially across individuals, making models trained on one subject difficult to transfer to another without calibration.

This report investigates this cross-subject generalization problem using the nine-subject BCI Competition IV-2a dataset, which contains four motor-imagery classes: left hand, right hand, feet, and tongue. Three modelling approaches are evaluated. First, a hybrid feature-engineering pipeline based on Wavelet Packet Decomposition (WPD), Filter-Bank Common Spatial Patterns (CSP), and a Multi-Layer Perceptron (MLP) is trained on a single subject and then transferred directly to the remaining subjects. This experiment exposes a substantial generalization gap, with an accuracy of 0.759 on the held-out trials of the training subject compared with 0.317 ± 0.074 on the eight unseen subjects.

Second, the same hybrid modelling approach is evaluated using Leave-One-Subject-Out (LOSO) cross-validation across all nine subjects, with Euclidean Alignment included to reduce subject-specific differences in covariance structure. This raises the mean cross-subject accuracy to 0.401 ± 0.125. Third, EEGNet, a compact convolutional neural network designed for EEG decoding, is trained end-to-end under the same LOSO protocol. Despite a restricted training budget imposed by CPU-only computation, EEGNet achieves the highest zero-shot cross-subject accuracy of the three approaches, reaching 0.439 ± 0.140 with a mean Cohen's κ of 0.252. A brief subject-specific fine-tuning stage further increases the mean accuracy to 0.476.

The experiments also include a pairwise transfer analysis to quantify the domain shift more directly. Mean within-subject accuracy is 0.656, whereas direct cross-subject transfer achieves only 0.254, close to the four-class chance level of 0.25. Taken together, the results show that cross-subject transfer remains a central challenge for MI-EEG classification. They also suggest that learning from multiple subjects and using end-to-end representations can improve zero-calibration performance, while a small amount of subject-specific calibration can provide a further improvement.

Keywords: EEG; Brain–Computer Interface; Motor Imagery; Cross-Subject Generalization; Common Spatial Patterns; Wavelet Packet Decomposition; EEGNet; Leave-One-Subject-Out; Transfer Learning.

Contents

1 Introduction

1.1 Motivation

Brain–Computer Interfaces (BCIs) aim to translate neural activity into control commands, providing a direct communication pathway between the brain and external devices without relying on the peripheral nervous system. Among non-invasive recording techniques, electroencephalography (EEG) is one of the most widely used because it is relatively inexpensive, portable, and provides high temporal resolution. Motor imagery (MI), which refers to mentally rehearsing a movement without actually performing it, is one of the most studied EEG paradigms. Imagined movements can alter sensorimotor rhythms, particularly the mu (8–13 Hz) and beta (13–30 Hz) bands, through event-related desynchronization and synchronization (ERD/ERS) [1]. Reliable decoding of these patterns has potential applications in assistive technologies for people with paralysis, stroke rehabilitation, and hands-free device control.

1.2 Challenges in EEG-Based Motor Imagery Classification

Despite extensive research, accurately classifying motor-imagery EEG remains difficult. Several factors contribute to this challenge:

- **Low signal-to-noise ratio.** The neural activity of interest is strongly attenuated as it passes through the brain, skull, and scalp, and EEG recordings are easily affected by ocular, muscular, and cardiac artifacts.
- **Non-stationarity.** EEG characteristics can change even within the same recording session. Factors such as fatigue, attention, and changes in electrode impedance can all affect the recorded signal.
- **High dimensionality with limited data.** A single trial may contain dozens of channels sampled at hundreds of Hz, while only a few hundred labelled trials are typically available for each subject. This creates a difficult balance between the number of features and the amount of training data.
- **Inter-subject variability.** Perhaps the most important challenge for practical BCI systems is that EEG patterns differ considerably across individuals. Differences in cortical anatomy, skull properties, electrode placement, and even the way a person performs motor imagery can change the spatial and spectral characteristics of the recorded signal.

1.3 The Inter-Subject Variability Problem

The last challenge is the main focus of this report. A classifier trained on one person's EEG learns spatial patterns and decision boundaries that are often closely tied to that individual's anatomy and cognitive strategy. When the same classifier is applied to a new subject without any additional calibration, a setting often referred to as *zero-calibration* or *subject-independent* decoding, performance can drop substantially toward chance level.

This is a major obstacle for the development of practical "plug-and-play" BCIs. Although subject-specific calibration can improve performance, it also adds extra time and effort for the user, which can be particularly inconvenient in clinical applications. For this reason, an important question is not only whether a model can classify motor imagery accurately for a given subject, but also how well it can transfer to people it has never seen before. This report therefore focuses on three related questions: how large the cross-subject generalization gap is, what factors contribute to it, and which modelling strategies can reduce it.

1.4 Objectives and Contributions

This report presents three experiments, implemented as separate Jupyter notebooks and evaluated on the same nine-subject dataset:

1. **Experiment 1 – Subject-Dependent Baseline.** A hybrid WPD + Filter-Bank-CSP + MLP pipeline is trained and validated exclusively on Subject 1. The trained model is then applied directly to Subjects 1–9 without any refitting. This experiment provides a simple baseline for measuring how much performance is lost when a model is transferred from one subject to others.

2. **Experiment 2 – LOSO with the Baseline Algorithm.** The same hand-engineered feature pipeline and MLP classifier are evaluated using a proper Leave-One-Subject-Out cross-validation protocol across all nine subjects. Euclidean Alignment is also introduced as a signal-normalization step to reduce differences between subjects.

3. **Experiment 3 – Deep Learning via EEGNet with LOSO.** EEGNet [2], a compact convolutional neural network designed specifically for EEG decoding, is used in place of the hand-engineered feature pipeline. The model is evaluated using the same LOSO protocol, while also taking into account the computational limitations of the available hardware.

The report compares the performance of these three approaches, investigates the extent of the observed domain shift between subjects, and discusses how the computational constraints affected the EEGNet experiment. The results are then used to identify practical directions for improving cross-subject MI-EEG classification in future work.

1.5 Report Structure

The remainder of the report is organized as follows. Section 2 introduces the dataset used throughout the experiments. Section 3 describes the preprocessing, feature-extraction, and modelling steps, highlighting both the components shared by the three experiments and the parts that differ between them. Section 4 presents the experimental setup and discusses the computational constraints, particularly those affecting EEGNet. The results of all three experiments are presented and compared in Section 5. Section 6 then discusses the main findings and their implications, followed by future directions in Section **??**. Finally, Section 7 summarizes the main conclusions of the study.

2 Dataset Description

All three experiments are based on the **BCI Competition IV Dataset 2a** [3], a widely used public benchmark for motor-imagery EEG classification.

Table 1: Summary of the BCI Competition IV-2a dataset as used in this project.

Property	**Value**
Subjects	9 (A01T – A09T)
Classes	4 – Left Hand, Right Hand, Feet, Tongue
Trials per subject	288 (72 per class, balanced)
Raw channels	25 (22 EEG + 3 EOG)
Sampling rate	250 Hz (hardware band-pass 0.5–100 Hz)
Electrode layout	Extended international 10–20 system
Cue events used	`769` (left), `770` (right), `771` (feet), `772` (tongue)
Trial timing	Fixation cross at t=0 s, visual cue at t=2 s (1.25 s duration), motor imagery sustained until t=6 s

Each of the nine subjects contributes the same set of 288 trials, with 72 trials for each of the four classes. This class balance removes class imbalance as a potential source of bias and makes the results directly comparable with the 0.25 chance level expected for four equally represented classes.

3 Methodology

3.1 Preprocessing Pipeline

The three experiments use two main preprocessing pipelines. Experiments 1 and 2 rely on a more extensive, neurophysiologically motivated preprocessing pipeline, while Experiment 3 uses a simpler pipeline before passing the data to EEGNet. This difference reflects the fact that the hybrid approach depends heavily on carefully prepared inputs, whereas EEGNet is designed to learn useful representations directly from the EEG signal.

3.1.1 Hybrid Pipeline Preprocessing (Experiments 1 & 2)

The preprocessing pipeline for the hybrid approach is implemented using MNE-Python [4] and consists of the following steps:

1. **Channel setup.** Raw GDF channel names are converted to standard 10–20 labels (for example, `EEG-C3` becomes `C3`). The three EOG channels are explicitly marked as `eog`, and a standard 10–20 montage is attached for visualization and topographic analysis.

2. **Re-referencing.** An average reference is applied across the EEG channels.

3. **Ocular artifact removal (ICA).** Independent Component Analysis (FastICA) is used to identify and remove components associated with eye movements. ICA is fitted on a temporary 1–40 Hz high-pass filtered copy of the data, retaining enough components to explain 99% of the variance, following the standard MNE workflow. Components showing a correlation with the EOG channels above the threshold of 2.5 are automatically marked for removal. The identified components are then removed from the original, non-high-passed signal. Across the nine subjects, between 1 and 3 components were removed per subject as ocular artifacts (Table 2).

4. **Band-pass filtering.** A zero-phase FIR filter with a `firwin` design is used to retain the 8–30 Hz mu+beta sensorimotor rhythm range, where motor-imagery-related ERD/ERS effects are expected to be most prominent.

5. **Epoching.** Each trial is extracted from -0.5 to 4.0 s relative to cue onset. Baseline correction is then applied using the pre-cue interval $[-0.5, 0]$ s to reduce DC offsets and slow drifts in the signal.

6. **Cropping to the imagery period.** The resulting epochs are further cropped to $[0.5, 3.5]$ s, giving a 3-second window that focuses on the main motor-imagery period. This removes much of the initial visually evoked and reaction-time activity as well as the final part of the trial, where fatigue or movement termination may have a stronger influence. This choice follows common practice in FBCSP-based MI-EEG analysis [5].

Before feature extraction, an ERD/ERS sanity check is performed using the mean PSD at C3 and C4 for left- and right-hand trials from Subject A01T. The resulting patterns show the expected contralateral desynchronization in the mu+beta range, providing a basic check that the preprocessing pipeline is preserving the relevant sensorimotor activity. The result is shown in Figure 10 (Appendix A).

Table 2: Per-subject ICA-based ocular artifact removal summary (Experiment 2 / LOSO preprocessing).

Subject	S01	S02	S03	S04	S05	S06	S07	S08	S09
ICA comps. removed	2	1	2	3	2	2	2	2	2
Trials	288	288	288	288	288	288	288	288	288

3.1.2 Cross-Subject Signal Alignment (Experiment 2 only)

Experiment 2 includes an additional **Euclidean Alignment** (EA) step [6] to reduce differences in the signal statistics across subjects. The alignment is performed independently for each subject before the LOSO split. First, the covariance matrix of each trial is computed and the mean covariance is obtained for that subject. Each trial is then whitened using the inverse square root of this subject-specific mean covariance.

Importantly, the alignment matrix is estimated using only the EEG trials of the corresponding subject. It does not use class labels or information from any other subject. As a result, applying EA before the LOSO split does not introduce cross-subject information leakage. The main purpose of this step is to make the second-order covariance structure of the recordings more comparable across subjects, since differences in covariance statistics are an important source of subject-to-subject variation in motor-imagery EEG.

3.1.3 EEGNet Pipeline Preprocessing (Experiment 3)

The EEGNet pipeline uses a simpler preprocessing scheme than the hybrid approach. This is consistent with the idea behind end-to-end deep learning models: rather than relying heavily on manually designed spatial and spectral features, the network is given a relatively broad representation of the signal and learns useful filters directly from the data.

1. **Re-referencing.** Common Average Reference (CAR) is applied to the EEG channels.

2. **Band-pass filtering.** A wider 4–38 Hz FIR filter is used instead of the 8–30 Hz mu+beta band used in the hybrid pipeline. The broader range preserves both lower- and higher-frequency information that the learned convolutional filters may be able to use.

3. **Epoching.** A 0–4.0 s window relative to cue onset is extracted for each trial, without applying explicit baseline correction.

4. **Resampling.** The epochs are downsampled from 250 Hz to 128 Hz, resulting in 513 samples per trial. This reduces the temporal size of the input and, in turn, lowers the computational cost of training the convolutional network.

5. **Per-trial normalization.** Each trial is independently z-scored on a channel-by-channel basis using the mean and standard deviation calculated from that trial. The resulting values are then clipped to ± 6 standard deviations. This step helps reduce differences in signal amplitude and scale between subjects, which can be particularly important in a LOSO setting. Since the normalization uses only statistics from the individual trial and does not rely on information from other trials or subjects, it does not introduce data leakage.

The BCI-IV-2a dataset provides two session types: `T`, containing the labelled training trials, and `E`, containing the evaluation trials. For Experiment 3, only the `T` sessions could be used. The true labels for the `E` sessions are provided separately as `.mat` files, which were not available in the working environment. As a result, the data loader automatically skipped those sessions, leaving 288 trials per subject, the same number used in Experiments 1 and 2.

3.2 Feature Engineering: Hybrid WPD + Filter-Bank CSP

Experiments 1 and 2 use the same general family of hand-engineered features. The approach combines Wavelet Packet Decomposition (WPD) for time-frequency analysis with Common Spatial Patterns (CSP) for spatial filtering, following the general idea of a Filter-Bank CSP (FBCSP) architecture [5].

3.2.1 Wavelet Packet Decomposition (WPD)

Unlike the standard Discrete Wavelet Transform, which recursively decomposes only the low-frequency branch, Wavelet Packet Decomposition recursively splits both the low- and high-frequency branches at each level. This produces a complete set of evenly spaced sub-bands. In this project, the Daubechies-4 (`db4`) wavelet is used. It is a common choice for EEG analysis because it provides a useful balance between smoothness and compact support.

Given the signal's Nyquist range of 0–125 Hz, a decomposition at level L produces 2^L sub-bands of equal width. Only the leaf nodes whose frequency ranges overlap the 8–30 Hz band-pass range are retained for feature extraction. In this way, WPD provides a finer time-frequency decomposition of the sensorimotor frequencies that were already selected during preprocessing.

- **Experiment 1** (Subject 1 only) uses $L = 4$, producing 16 leaf nodes with a bandwidth of 7.8125 Hz each. Three of these overlap the target frequency range: `aaad` (7.81–15.63 Hz), `aadd` (15.63–23.44 Hz), and `aada` (23.44–31.25 Hz).

- **Experiment 2** (LOSO) uses a deeper decomposition with $L = 5$, giving 32 leaf nodes with a bandwidth of 3.91 Hz each. Six relevant sub-bands are retained: `aaadd`, `aaada`, `aadda`, `aaddd`, `aadad`, and `aadaa`, covering the range from 7.81 to 31.25 Hz.

For each selected sub-band, two types of features are extracted for every channel and trial:

1. **Statistical features.** Energy, variance, and Shannon entropy are calculated from the wavelet coefficients. These features are extracted only from a 5-channel sensorimotor region of interest (ROI): {C3, C1, Cz, C2, C4}.
2. **Sub-band reconstruction.** The coefficients of the selected leaf node are inverse-transformed while all other leaf nodes are set to zero. This reconstructs a band-limited version of the original multi-channel signal, which is then passed to CSP.

3.2.2 Filter-Bank Common Spatial Patterns (CSP)

A separate CSP model [7] is fitted for each WPD sub-band. Ledoit–Wolf shrinkage regularization is used during CSP fitting, and the resulting log-variance features are retained. The purpose of CSP is to find spatial filters that emphasize differences in signal variance between the motor-imagery classes for each sub-band.

This "CSP-per-WPD-subband" setup follows the general Filter-Bank CSP idea introduced by Ang et al. [5], but uses WPD as the filter bank instead of a set of manually selected band-pass filters. This allows the frequency decomposition to be defined more finely by the wavelet packet structure.

3.2.3 Feature Fusion and Leakage Prevention

For each trial, the log-variance features produced by CSP across all selected sub-bands are concatenated with the WPD statistical features extracted from the sensorimotor ROI. The complete feature vector is then standardized to zero mean and unit variance.

Because the CSP spatial filters depend on the class labels and the standardization step depends on the data distribution, both are fitted only on the training portion of each split. In Experiment 1, this means the training set from the single-subject train/validation/test split. In Experiment 2, they are fitted using only the pooled eight-subject training fold in each LOSO iteration. Once fitted, these transformations are applied to the validation, test, or held-out subject data without being refitted. This prevents information from the evaluation data from leaking into the training process.

Table 3: Final hybrid feature vector composition.

Experiment	CSP comp.	Sub-bands	CSP feat.	WPD stat. feat.	**Total**
1 (Subject-dependent)	4	3	$4 \times 3 = 12$	$5 \times 3 \times 3 = 45$	**57**
2 (LOSO)	8	6	$8 \times 6 = 48$	$5 \times 6 \times 3 = 90$	**138**

The final feature vector is therefore considerably larger in Experiment 2, with 138 features compared with 57 in Experiment 1. This difference comes from the deeper WPD decomposition and the larger number of CSP components used for each sub-band in the LOSO experiment. The increased feature dimensionality also motivated the use of a larger weight-decay coefficient for the MLP described in Section 3.3.1.

3.3 Classification Models

3.3.1 Multi-Layer Perceptron (Experiments 1 & 2)

Both experiments based on the hybrid features use the same compact feed-forward Multi-Layer Perceptron (MLP):

$$\text{Input} \to \text{Linear}(\cdot, 128) \to \text{BatchNorm} \to \text{ReLU} \to \text{Dropout} \to \text{Linear}(128, 64) \to \text{BatchNorm} \to \text{ReLU} \to \text{Dropout} \to \text{Linear}(64, 4)$$

The network is trained using the Adam optimizer and cross-entropy loss. Early stopping based on validation loss is used in Experiment 1, while Experiment 2 follows a short fixed training schedule with early stopping. In addition to the MLP, a simple multinomial Logistic Regression classifier is trained on the same hybrid features in both experiments. This provides a useful lower-complexity baseline and helps determine whether the additional non-linearity of the MLP provides a meaningful benefit.

3.3.2 EEGNet (Experiment 3)

EEGNet [2] is a compact convolutional neural network designed specifically for EEG decoding. Its architecture consists of three main stages:

1. **Temporal convolution.** A 2-D convolution with F_1 filters of length `kernel_length` is applied along the time axis. These filters learn temporal patterns at different frequencies and therefore serve a role similar to a learned band-pass filter bank.

2. **Depthwise spatial convolution.** A depthwise convolution spans all EEG channels (kernel size $= n_{\text{channels}} \times 1$) with depth multiplier D. This layer learns spatial filters that are conceptually similar to CSP filters, with each spatial filter operating on the features produced by an individual temporal filter. Batch normalization, ELU activation, average pooling, and dropout are then applied.

3. **Separable temporal convolution.** A depthwise convolution followed by a pointwise convolution further processes the resulting $F_1 \times D$ feature maps and combines them into F_2 output feature maps. This block is followed by another batch normalization, ELU activation, average pooling, and dropout stage.

The resulting feature maps are flattened and passed to a final fully connected layer that produces the logits for the four classes. Following the original EEGNet design, max-norm constraints are also applied after every optimizer step to two sets of weights: the depthwise-convolution kernel (max norm 1.0) and the final dense layer (max norm 0.25). These constraints provide an additional form of regularization, which is particularly useful given the relatively limited amount of training data available for each subject.

Table 4: EEGNet architectural hyperparameters used in Experiment 3.

Hyperparameter	**Value**
Temporal filters (F_1)	8
Depth multiplier (D)	2
Pointwise filters (F_2)	16
Temporal kernel length	64 samples
Dropout rate	0.5
Depthwise max-norm	1.0
Dense max-norm	0.25
Input shape	$(1, 22, 513)$ – (channel-dim, EEG channels, time samples)
Output classes	4

3.4 Validation Strategies

3.4.1 Subject-Dependent Single-Subject Evaluation (Experiment 1)

For Experiment 1, the complete pipeline is fitted using only the data from Subject A01T. This includes the WPD sub-band selection, CSP spatial filters, feature scaler, and MLP weights. The subject's 288 trials are divided using a stratified 60%/20%/20% train/validation/test split, resulting in 172, 58, and 58 trials, respectively.

Once trained, the complete pipeline is applied to Subjects A02T–A09T without any additional training or refitting. Every trial from these subjects is treated as a test trial because none of their data is used during model fitting. This setup provides a direct measurement of how well a model trained for one subject transfers to completely unseen subjects, and therefore gives a simple estimate of the zero-calibration cross-subject generalization gap.

3.4.2 Leave-One-Subject-Out Cross-Validation (Experiments 2 & 3)

Experiments 2 and 3 use Leave-One-Subject-Out (LOSO) cross-validation. The procedure is repeated nine times, once for each subject. In each fold, one subject is kept entirely separate as the test set, while the remaining eight subjects are pooled together for training. A stratified 15% subset of this training pool is further reserved for validation and, where applicable, early stopping.

The model or feature-extraction pipeline is fitted from scratch for each fold, and the held-out subject is not used at any point during training or model selection. This setup provides a more reliable estimate of subject-independent, or zero-calibration, performance while still allowing every subject to serve as the test subject once.

3.5 Evaluation Metrics

The experiments are evaluated using several complementary metrics: **accuracy**, **per-class precision, recall, and F1-score**, and **macro-averaged F1-score**. Confusion matrices are also reported both as raw class counts and as row-normalized matrices to make differences in class-wise performance easier to interpret.

For Experiment 3, **Cohen's kappa** [8] is also reported. Kappa measures agreement beyond what would be expected by chance and is commonly used in BCI research, where accuracy alone can sometimes give an incomplete picture of performance. It is defined as

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o is the observed accuracy and p_e is the accuracy expected by chance based on the class marginals. Since all four classes in this dataset are perfectly balanced, the chance level used throughout the report is $p_e = 0.25$.

4 Experimental Setup and Computational Constraints

4.1 Software and Hardware Environment

All three notebooks were implemented in Python. MNE-Python [4] was used for EEG input/output and preprocessing, PyWavelets for Wavelet Packet Decomposition, scikit-learn for CSP, Logistic Regression, and evaluation metrics, and PyTorch [9] for both the MLP and EEGNet models.

A practical limitation of the experiments is that all three notebooks were run on CPU only, since no CUDA-capable GPU was available in the working environment. This limitation had the greatest effect on Experiment 3 and directly influenced the training budget chosen for EEGNet.

4.2 Hyperparameter Summary Across Experiments

Table 5 summarizes the main preprocessing choices, model settings, and training hyperparameters used in the three experiments.

Table 5: Consolidated hyperparameter comparison across the three experiments.

Setting	Exp. 1 (Subj.-dep.)	Exp. 2 (LOSO hybrid)	Exp. 3 (EEGNet LOSO)
Band-pass filter	8–30 Hz	8–30 Hz	4–38 Hz
Epoch window (post-cue)	0.5–3.5 s	0.5–3.5 s	0–4.0 s
Sampling rate	250 Hz	250 Hz	128 Hz (resampled)
Cross-subject alignment	–	Euclidean Alignment	Per-trial z-score
WPD wavelet / level	`db4` / 4	`db4` / 5	–
Relevant sub-bands	3	6	–
CSP components / sub-band	4	8	–
Feature vector length	57	138	(end-to-end, no hand features)
Classifier	MLP (57-128-64-4)	MLP (138-128-64-4)	EEGNet (F_1=8, D=2, F_2=16)
Optimizer	Adam	Adam	Adam
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	2×10^{-4}	10^{-4}
Batch size	16	32	128
Max. epochs	150	25	42
Early-stop patience	30 (val. loss)	5 (val. loss)	6 (val. acc.)
Validation split	Stratified 20%	15% of pooled train	15% of pooled train
Data augmentation	–	–	Disabled (available: noise + time-shift)
Label smoothing	–	–	0.05

4.3 Computational Constraints on EEGNet Training

The EEGNet experiment was run under a deliberately limited training budget because of the available CPU-only hardware. The notebook defines `EPOCHS=42` and `PATIENCE=6` as a fast configuration, with the explicit trade-off that shorter training can reduce computation at the cost of some accuracy. The notebook also includes a more extensive configuration with `EPOCHS=300` and `PATIENCE=30`, together with data augmentation, intended to give the model more opportunity to learn.

In practice, the reduced 42-epoch, 9-fold LOSO experiment took more than one hour to run on CPU. The full 300-epoch configuration was estimated to require approximately seven hours or more on the same hardware and was therefore not practical within the available project budget.

The effect of this limitation can also be seen in the individual LOSO folds. Some folds reached the maximum allowed training budget without clearly plateauing; for example, Subjects 01 and 02 trained for the full 42 epochs. Other folds stopped earlier, such as Subject 08 at epoch 22, while Subject 05 reached epoch 42. These differences suggest that the shorter training schedule did not give every fold enough opportunity to fully optimize the model.

For this reason, the EEGNet results reported in this study should be interpreted as the performance of a resource-constrained training configuration rather than as a measure of the model's full potential. A longer training schedule, together with the available augmentation strategy, could potentially improve cross-subject performance. This limitation is important when interpreting the comparison with the other approaches and is discussed further in Section 6.

5 Results

5.1 Experiment 1: Subject-Dependent Baseline

5.1.1 In-Sample Performance (Subject A01T)

As a first check, a Logistic Regression classifier was trained on the 57-dimensional hybrid feature vector. It achieved a 5-fold cross-validated training accuracy of 0.739 ± 0.039 and a held-out test accuracy of 0.793. This suggests that the extracted features contain a substantial amount of class-discriminative information for Subject 1 and provides a useful baseline before evaluating the MLP.

The MLP was trained for 65 epochs before early stopping, with the weights corresponding to the lowest validation loss restored. On the held-out 20% test split of Subject A01T, the model achieved an **overall accuracy of** 0.759 and a macro-F1 score of 0.760. The detailed per-class results are shown in Table 6, and the corresponding confusion matrix is presented in Figure 1.

Table 6: Experiment 1 – per-class test performance, Subject A01T (in-sample, held-out split).

Class	Precision	Recall	F1-Score	Support
Left Hand	1.000	0.667	0.800	15
Right Hand	0.722	0.929	0.812	14
Feet	0.632	0.800	0.706	15
Tongue	0.818	0.643	0.720	14
Macro Avg.	**0.793**	**0.760**	**0.760**	58

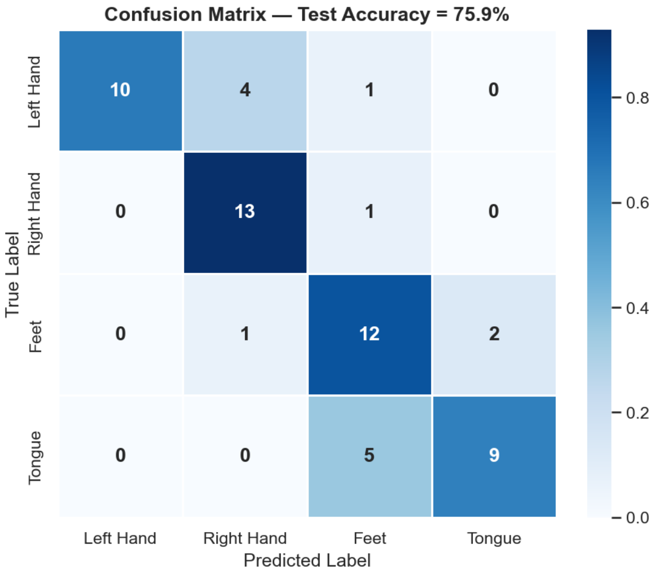


Figure 1: Confusion matrix for the MLP classifier evaluated on the held-out Subject A01T test split (overall accuracy = 75.9%).

5.1.2 Cross-Subject Generalization (Zero-Calibration Test on All Nine Subjects)

After fitting the complete pipeline on Subject 1, including the WPD sub-band selection, CSP filters, feature scaler, and MLP weights, the model was applied directly to Subjects A01T–A09T. No part of the pipeline was refitted for the other subjects. The resulting accuracy and macro-F1 scores are summarized in Table 7.

Table 7: Experiment 1 – cross-subject generalization of the Subject-1-trained pipeline. A01T is the trained (in-sample) subject; A02T–A09T are entirely unseen.

Subject	N Trials	Accuracy	Macro F1	Status
A01T	58	0.759	0.760	Trained (in-sample)
A02T	288	0.316	0.254	Unseen
A03T	288	0.431	0.310	Unseen
A04T	288	0.326	0.296	Unseen
A05T	288	0.274	0.174	Unseen
A06T	288	0.253	0.113	Unseen
A07T	288	0.424	0.372	Unseen
A08T	288	0.257	0.115	Unseen
A09T	288	0.253	0.108	Unseen
Mean, all 9 subjects		0.366 ± 0.163		–
Mean, unseen subjects (A02T–A09T)		0.317 ± 0.074	0.218	–
Chance level		0.250		–

The resulting **generalization gap** is large. The model achieves 0.759 accuracy on the held-out trials from its training subject, but only 0.317 ± 0.074 on the eight unseen subjects. The difference is therefore

$$0.759 - 0.317 = 0.442,$$

corresponding to a loss of roughly *44 percentage points* when the model is transferred to new individuals.

The average performance on the unseen subjects is still slightly above the four-class chance level of 0.25, but the individual results vary considerably. Accuracy ranges from 0.253 for Subjects A06T and A09T, which is essentially at chance, to 0.431 for Subject A03T. This spread shows that cross-subject transfer is not equally difficult for all subjects and that substantial differences in decodability remain even within the unseen group.

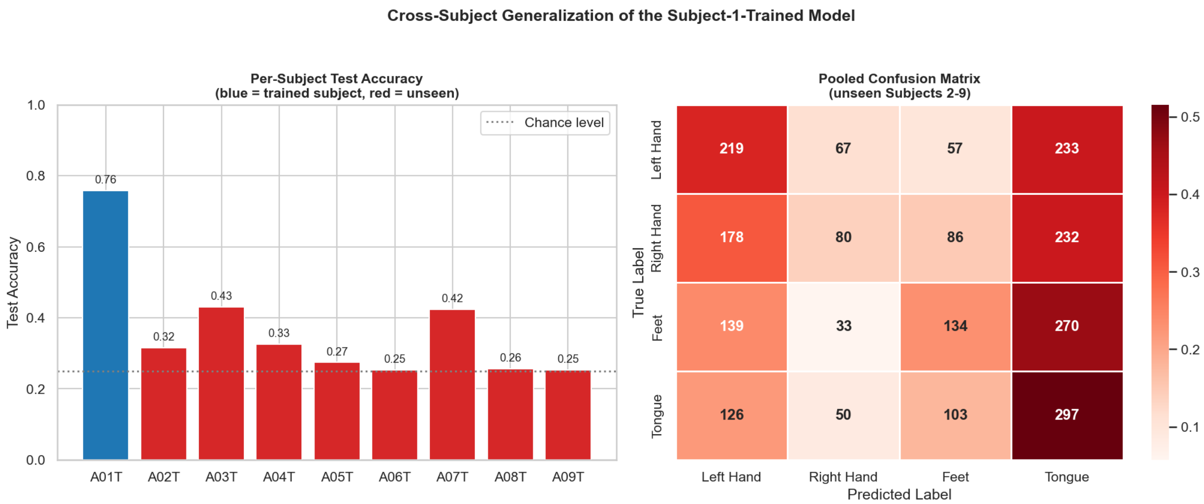


Figure 2: Left: per-subject test accuracy of the Subject-1-trained pipeline (blue = trained subject, red = unseen subjects), with the chance-level reference line. Right: pooled confusion matrix across the eight unseen subjects (A02T–A09T).

5.2 Experiment 2: Leave-One-Subject-Out with the Baseline Algorithm

Table 8 summarizes the results of the LOSO experiment for both the Hybrid WPD-CSP-MLP model and the CSP+Logistic-Regression baseline. This experiment uses the larger 138-dimensional feature vector, the deeper WPD decomposition, and Euclidean Alignment described in Section 3.

Table 8: Experiment 2 – LOSO test results per held-out subject.

Test Subject	MLP Accuracy	Baseline (LogReg) Accuracy	Macro F1 (MLP)
S01	0.566	0.503	0.511
S02	0.247	0.285	0.157
S03	0.524	0.493	0.504
S04	0.347	0.306	0.294
S05	0.295	0.285	0.229
S06	0.285	0.333	0.183
S07	0.351	0.285	0.298
S08	0.569	0.524	0.568
S09	0.424	0.420	0.379
Mean ± SD	**0.401 ± 0.125**	0.382 ± 0.103	0.347 ± 0.152
Chance level		0.250	

The Hybrid WPD-CSP-MLP model achieves a mean LOSO accuracy of 0.401 ± 0.125, compared with 0.382 ± 0.103 for the CSP+Logistic-Regression baseline. The difference is relatively small, but the MLP also performs better than the mean unseen-subject accuracy obtained in Experiment 1 (0.317 ± 0.074).

One likely reason for this improvement is that the LOSO model is trained using data from eight different subjects in each fold. Instead of learning primarily from the characteristics of a single individual, the model is exposed to a wider range of EEG patterns during training. This gives the learned spatial filters and decision boundaries a better chance of handling the variability encountered in the held-out subject, even though the experiment does not use an explicit domain-adaptation method beyond Euclidean Alignment.

There is still considerable variation between folds. Subject S08 is the easiest subject to generalize to, with an accuracy of 0.569, whereas S02 is the most difficult, with an accuracy of 0.247, which is essentially the four-class chance level. A similar pattern of subject-dependent difficulty also appears in Experiment 3.

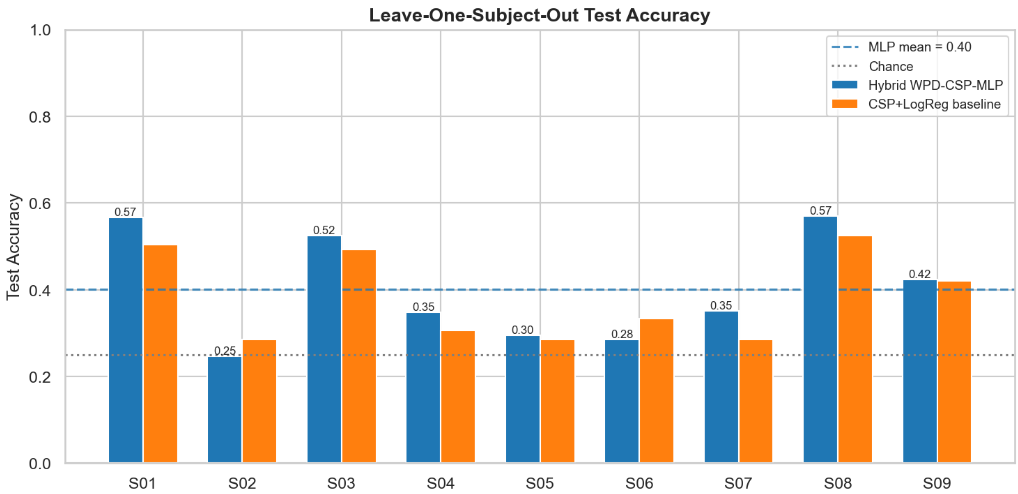


Figure 3: LOSO test accuracy per held-out subject for the Hybrid WPD-CSP-MLP classifier compared against the CSP+Logistic-Regression baseline, with the LOSO-MLP mean and chance-level reference lines.

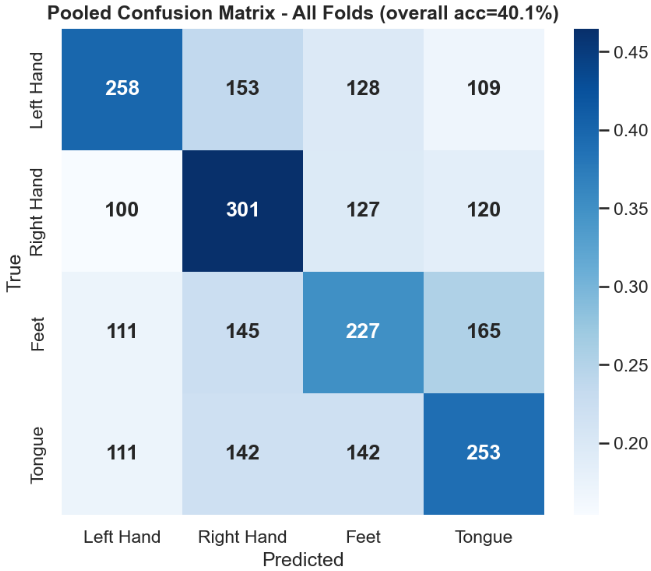


Figure 4: Pooled confusion matrix across all nine LOSO folds (Hybrid WPD-CSP-MLP), row-normalized with raw counts annotated.

5.2.1 Quantifying the Domain Shift: Pairwise Transfer Analysis

To get a more direct picture of the inter-subject domain shift, an additional 9×9 pairwise transfer experiment was performed using the faster CSP+Logistic-Regression pipeline. For every ordered pair of subjects (i, j), the model was trained on all available data from subject i and then tested on all data from subject j. The diagonal entries were handled differently: they represent within-subject held-out performance using a 70%/30% train/test split from the same subject. This provides a direct comparison between within-subject performance and subject-to-subject transfer.

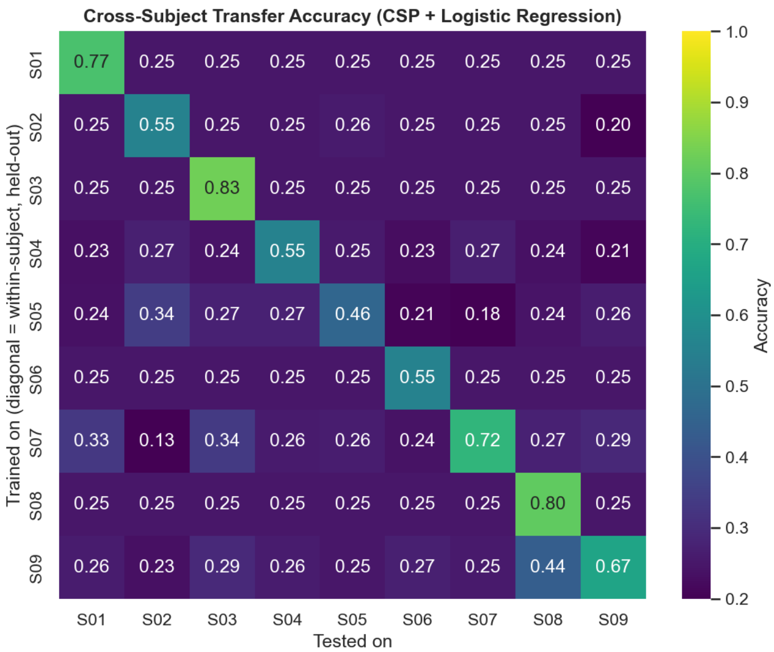


Figure 5: Pairwise cross-subject transfer accuracy matrix (CSP + Logistic Regression). Diagonal entries are within-subject held-out accuracy; off-diagonal entries are direct subject-to-subject transfer accuracy.

The difference between the two settings is substantial. The mean **within-subject** accuracy across the diagonal entries is **0.656**, whereas the mean **cross-subject** accuracy across the off-diagonal entries is only **0.254**, which is essentially the four-class chance level of 0.25.

This provides one of the clearest demonstrations in the study of the difficulty of transferring CSP-based models between subjects. Spatial filters learned from one subject's covariance structure do not transfer well to another subject's covariance structure when the model is trained on only one individual. The results also help explain why the approaches that either expose the model to multiple subjects during training, as in the LOSO experiments, or explicitly reduce subject-specific differences through alignment can perform better.

5.3 Experiment 3: Deep Learning via EEGNet with LOSO

Table 9: Experiment 3 – EEGNet LOSO test results per held-out subject, including subsequent subject-specific fine-tuning results (Section 5.3.1).

Test Subject	Zero-Shot Accuracy	Cohen's κ	Fine-Tuned Accuracy
S01	0.660	0.546	0.674
S02	0.299	0.065	0.319
S03	0.587	0.449	0.660
S04	0.403	0.204	0.424
S05	0.257	0.009	0.257
S06	0.351	0.134	0.424
S07	0.347	0.130	0.396
S08	0.497	0.329	0.569
S09	0.552	0.403	0.563
Mean ± SD	$\mathbf{0.439 \pm 0.140}$	0.252	**0.476**
CSP+MLP LOSO baseline (Exp. 2)	0.401 (rounded to 0.400 in-notebook)		
Chance level	0.250		

EEGNet achieves a **mean zero-shot LOSO accuracy of** 0.439 ± 0.140 and a **mean Cohen's** κ **of** 0.252. This is an absolute improvement of 0.039, or roughly 4 percentage points, over the Experiment 2 CSP+MLP LOSO result. The comparison is particularly interesting because EEGNet does not rely on the hand-engineered spatial and spectral features used by the hybrid pipeline. Instead, the network learns its temporal and spatial filters directly from the EEG data.

The improvement is not uniform across subjects. As in Experiment 2, some subjects are much easier to decode than others. Subject S01 achieves the highest EEGNet accuracy at 0.660, while S05 and S02 remain close to chance, with accuracies of 0.257 and 0.299, respectively. This subject-dependent pattern appears in both the hybrid and deep-learning approaches. The consistency suggests that at least part of the difficulty is related to differences in the recordings themselves, such as the strength of the motor-imagery-related signal or the amount of noise and artifact, rather than being caused solely by the choice of classifier.

The class-wise results also show a tendency for Left and Right Hand trials to be decoded more reliably than Feet and Tongue trials. This pattern is consistent with the stronger spatial lateralization that is often associated with hand motor imagery. The aggregated confusion matrix across the LOSO folds, although not shown here, reflects the same general pattern.

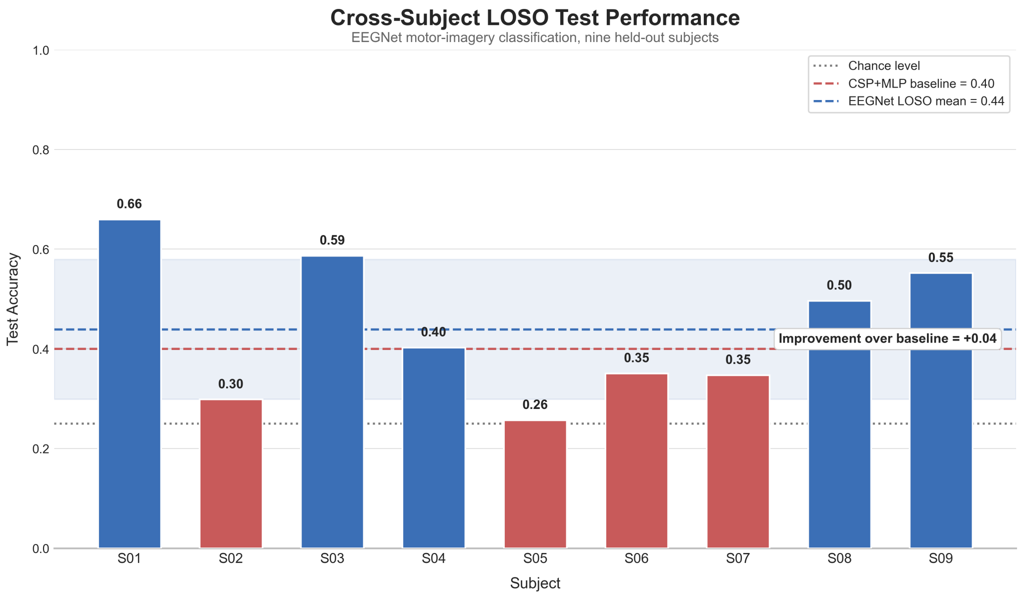


Figure 6: LOSO test accuracy per subject for EEGNet, with reference lines for the CSP+MLP LOSO baseline mean (40%) and chance level (25%), and the overall EEGNet mean ±1 SD band.

5.3.1 Subject-Specific Fine-Tuning (Calibration)

To examine a more realistic deployment scenario, each EEGNet model trained under LOSO was subsequently fine-tuned using half of the previously unseen target subject's data. This corresponds to a 50% calibration split, followed by 10 additional training epochs at a reduced learning rate. The fine-tuned model was then evaluated on the remaining 50% of that subject's trials. The zero-shot and fine-tuned results are reported in Table 9.

Fine-tuning increased the mean accuracy from 0.439 to **0.476**, an absolute improvement of 3.7 percentage points using only 144 calibration trials per subject. Every subject either improved or maintained its zero-shot accuracy. For example, S06 increased from 0.351 to 0.424, while S08 increased from 0.497 to 0.569. S05 was the only subject with no improvement, remaining at 0.257.

These results suggest that a relatively small amount of subject-specific calibration can be useful when a completely subject-independent model is not accurate enough. In this experiment, the calibration step helped narrow the gap between zero-shot transfer and subject-specific decoding without requiring the model to be retrained from scratch for the new subject.

5.4 Comparative Analysis Across All Three Experiments

Table 10: Summary comparison of cross-subject / LOSO mean accuracy across all experimental approaches.

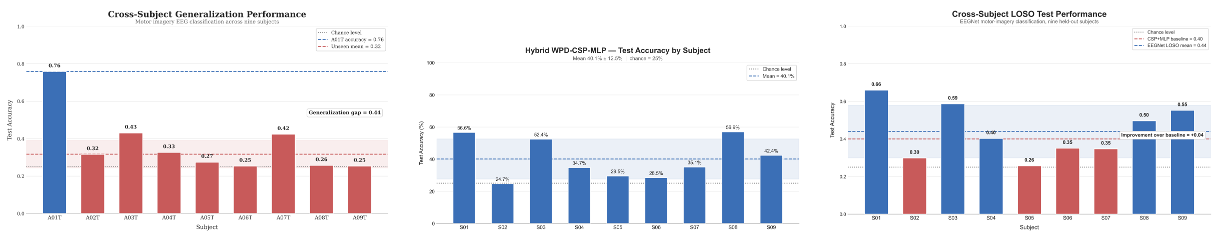
Approach	Mean Accuracy	Macro F1 / κ	Validation Scheme
Exp. 1: Subject-dependent (in-sample, A01T)	0.759	F1 = 0.760	Single-subject held-out split
Exp. 1: Subject-dependent (unseen, A02T–A09T)	0.317 ± 0.074	F1 = 0.218	Zero-calibration transfer
Exp. 2: LOSO, CSP+LogReg baseline	0.382 ± 0.103	–	LOSO (9-fold)
Exp. 2: LOSO, Hybrid WPD-CSP-MLP	0.401 ± 0.125	F1 = 0.347	LOSO (9-fold)
Exp. 3: LOSO, EEGNet (zero-shot)	**0.439 ± 0.140**	κ = 0.252	LOSO (9-fold)
Exp. 3: LOSO, EEGNet (fine-tuned)	**0.476**	–	LOSO + 50% calibration
Chance level (4 balanced classes)	0.250	–	–

The results in Table 10 show a clear progression across the three experiments. When the model is trained only on one subject, performance on unseen subjects drops to 0.317 ± 0.074. Training on multiple subjects through the LOSO setup improves the hybrid WPD-CSP-MLP result to 0.401 ± 0.125. Replacing the hand-engineered feature pipeline with EEGNet increases the zero-shot LOSO accuracy further to 0.439 ± 0.140. Finally, adding a short subject-specific calibration step raises the mean accuracy to 0.476.

Taken together, these results suggest that both broader exposure to inter-subject variability during training and learned representations can help with cross-subject MI-EEG decoding. The fine-tuning results also show that even limited access to data from the target subject can provide a further improvement. At the same time, the EEGNet result should be interpreted in light of the restricted training budget described in Section 4.3.

5.5 Side-by-Side Comparison of the Three Final Per-Subject Accuracy Charts

Each of the three notebooks ends with a summary chart showing the accuracy for all nine subjects. Figure 7 brings these three charts together in a single row, making it easier to compare the overall performance and the subject-level patterns across the experiments. A short description is provided below each panel to highlight the main results.



(a) Experiment 1 – Subject-Dependent

Naive pipeline trained on Subject A01T only, then tested on all nine subjects. A01T (in-sample) scores 0.759; the eight unseen subjects average 0.317 ± 0.074. Best unseen: A03T (0.431). Worst: A06T and A09T (tied, 0.253, at chance).

(b) Experiment 2 – LOSO Hybrid WPD-CSP-MLP

Leave-One-Subject-Out accuracy for the hand-engineered WPD+CSP+MLP pipeline, all nine subjects now pooled during training. Mean accuracy 0.401 ± 0.125. Best: S08 (0.569), closely followed by S01 (0.566). Worst: S02 (0.247, at chance).

(c) Experiment 3 – LOSO EEGNet

Leave-One-Subject-Out accuracy for the end-to-end EEGNet classifier, same protocol as (b) but with learned rather than hand-engineered features. Mean accuracy 0.439 ± 0.140 ($\kappa = 0.252$). Best: S01 (0.660). Worst: S05 (0.257, at chance), with S02 close behind (0.299).

Figure 7: The three end-of-notebook per-subject accuracy summary charts, arranged side by side, with a brief explanation directly beneath each: (a) the naive single-subject-trained pipeline, (b) the LOSO Hybrid WPD-CSP-MLP classifier, and (c) the LOSO EEGNet classifier. All three plot the same nine subjects, allowing a direct visual comparison of both the overall accuracy level and the subject-to-subject pattern across approaches.

5.5.1 Overall Results

Looking at the three panels together makes the progression across the experiments clear. In panel (a), performance on the unseen subjects remains close to the 0.25 chance level, with a mean accuracy of 0.317. In panel (b), the bars generally move upward after multiple subjects are included during training, giving a mean LOSO accuracy of 0.401. Panel (c) shows a further increase with EEGNet, reaching a mean accuracy of 0.439.

The improvement is therefore not limited to a single subject. Instead, the overall level of performance shifts upward from the single-subject transfer setting to LOSO training and then to the EEGNet-based approach. Several subjects, particularly S01, S03, and S09, reach accuracies above 0.55 with EEGNet, whereas the hand-engineered pipelines generally remain below this level except for the in-sample A01T result in panel (a).

5.5.2 Best- and Worst-Performing Subjects

The three panels also reveal a noticeable pattern at the subject level. **Subject 01** is consistently one of the easiest subjects to decode. It achieves the highest in-sample accuracy in panel (a) at 0.759, ranks among the strongest subjects in panel (b), and reaches the highest accuracy in panel (c) at 0.660. **Subject 03** shows a similar pattern: it is the best-performing unseen subject in Experiment 1 and remains among the stronger subjects in both LOSO experiments.

At the other end of the range, **Subject 02** performs poorly across all three settings: 0.316 in the unseen-subject evaluation of Experiment 1, 0.247 in the hybrid LOSO experiment, and 0.299 with EEGNet. **Subject 05** shows a similarly weak pattern, with accuracies of 0.274, 0.295, and 0.257, respectively.

The fact that these subjects remain difficult across different preprocessing and modelling choices suggests that the observed performance differences are not explained by a single classifier alone. Differences in signal quality, the strength of the motor-imagery response, or the amount of subject-specific noise may be contributing to the difficulty. However, the present experiments do not isolate the individual contribution of these factors, so this interpretation should be treated as a plausible explanation rather than a definitive one.

5.5.3 General Performance Pattern

An additional point becomes visible when considering the spread of the results rather than only their mean. The standard deviation across subjects increases from panel (a) to panel (c), from 0.074 to 0.125 and then to 0.140, even though the mean accuracy improves.

This means that the gains from moving to LOSO training and then to EEGNet are not distributed equally across subjects. Subjects that are already relatively easy to decode, such as S01, S03, S08, and S09, tend to benefit more from the stronger approaches. In contrast, the most difficult subjects, particularly S02 and S05, remain close to the 0.25 chance level.

This pattern is consistent with the pairwise-transfer results in Section 5.2 (Figure 5), which also showed a large difference between within-subject and cross-subject performance. Taken together, the results suggest that part of the cross-subject gap is related to persistent differences between individuals that are not fully removed simply by changing the feature representation or classifier.

6 Discussion

6.1 Domain Shift Across Subjects

The pairwise transfer-accuracy heatmap in Figure 5 provides a clear view of the domain shift present in this dataset. The mean within-subject accuracy is 0.656, whereas direct cross-subject transfer reaches only 0.254, which is essentially the four-class chance level.

One important reason is that CSP learns spatial filters from the covariance structure of the subject used for training. That structure can differ substantially across individuals because of factors such as skull and scalp conductivity, cortical anatomy, electrode placement, and differences in how subjects perform motor imagery. A spatial pattern that is useful for one subject may therefore be much less useful for another.

The WPD-based statistical features can be affected by similar differences. For example, the distribution of signal energy across the mu and beta ranges may vary between subjects, meaning that a feature that is informative for one individual is not necessarily equally informative for another. The pairwise results therefore provide a useful explanation for why a model that works well within one subject can perform close to chance when transferred directly to someone else.

6.2 Why LOSO Training Helps, But Only Modestly

In Experiment 2, the model is trained using data from eight subjects rather than a single individual. This exposes the feature-extraction and classification stages to a wider range of EEG patterns before they are evaluated on the held-out subject. The improvement from 0.317 in the single-subject transfer experiment to 0.401 with LOSO is consistent with this effect.

However, the improvement is still limited. Training on several subjects does not remove the underlying differences between them; instead, the learned model has to find features and decision boundaries that work reasonably well across a heterogeneous group. In that sense, the result is a compromise across subjects rather than a representation explicitly adapted to the distribution of the held-out individual.

This also helps explain why Euclidean Alignment alone does not close the entire performance gap. It reduces part of the subject-specific variation in the covariance structure, but it does not account for every difference between subjects or for differences in the discriminative patterns themselves.

6.3 Why Deep Representations (EEGNet) Generalize Better

EEGNet performs somewhat better than the hybrid pipeline in the LOSO setting, although the difference is not large. There are several reasonable explanations for this result:

- **Wider, jointly learned temporal filters.** EEGNet learns $F_1 = 8$ temporal filters directly from the 4–38 Hz input rather than being restricted to the manually selected 8–30 Hz range used in the hybrid pipeline. This gives the model the opportunity to use frequency information outside the narrower mu+beta band when it is helpful for classification.

- **End-to-end optimization.** The temporal filters, spatial filters, and classifier are optimized together using a single loss function. In contrast, the hybrid pipeline constructs WPD and CSP features before sending them to the MLP. The end-to-end setup allows EEGNet to adjust its learned representation in response to the final classification objective.

- **Built-in regularization.** The max-norm constraints applied to the depthwise convolution and dense layer provide an additional form of regularization. Combined with dropout, these constraints may help control overfitting given the relatively limited amount of labelled data.

- **Per-trial normalization.** EEGNet normalizes each trial independently rather than applying a scaler estimated from the training set. This removes part of the amplitude and scale variation before the signal reaches the learned filters and may therefore make the input representation more consistent across subjects.

These factors are plausible contributors to the observed improvement, but the experiments were not designed to isolate their individual effects. The difference between the hybrid pipeline and EEGNet should therefore be interpreted as an empirical comparison of the complete pipelines, rather than evidence that any single EEGNet component is responsible for the improvement.

6.4 Impact of the Computational Constraint on the EEGNet Results

The EEGNet results should be interpreted in the context of the limited training budget described in Section 4.3. The model was trained for at most 42 epochs with an early-stopping patience of 6, rather than the longer 300-epoch configuration considered in the notebook.

Several folds reached the maximum epoch budget without clear evidence of convergence. In particular, S01 and S02 trained for the full 42 epochs rather than stopping earlier because of the patience criterion. This indicates that, for at least some subjects, the available training budget may not have been sufficient to fully optimize the model.

It is therefore reasonable to view the reported EEGNet accuracy as a result obtained under a resource-constrained training setup rather than as the maximum performance that the architecture could achieve on this dataset. Longer training, together with the available augmentation strategy, could potentially improve the results. However, such an improvement was not directly measured in this study, so it should be treated as a future possibility rather than a demonstrated outcome. The computational limitation is consequently an important part of the context when comparing EEGNet with the other approaches used in this report.

7 Conclusion

This report examined three different approaches to cross-subject motor-imagery EEG classification using the nine-subject BCI Competition IV-2a dataset. The first experiment showed that a hybrid WPD-CSP-MLP pipeline can perform reasonably well when trained and tested within the same subject, achieving an accuracy of 0.759 on the held-out A01T trials. However, when the same model was applied directly to previously unseen subjects, the mean accuracy dropped to 0.317 ± 0.074, with several subjects performing close to the 0.25 chance level. This large gap highlights how difficult it is to transfer a model trained on one person's EEG to another person without calibration.

The second experiment addressed this problem using Leave-One-Subject-Out training. By exposing the model to data from multiple subjects during training and adding Euclidean Alignment, the mean accuracy increased to 0.401 ± 0.125. The improvement was meaningful but still limited, suggesting that simply pooling data from multiple subjects does not completely remove the differences between their EEG signals.

EEGNet performed better in the third experiment, reaching a mean zero-shot LOSO accuracy of 0.439 ± 0.140 with a Cohen's κ of 0.252. Despite the restricted training budget caused by CPU-only computation, it achieved the highest zero-shot performance among the three approaches. A short subject-specific fine-tuning stage increased the mean accuracy further to 0.476, showing that even a relatively small amount of calibration data can improve performance on a previously unseen subject.

Overall, the experiments point to two main observations. First, cross-subject variability is a major challenge for MI-EEG classification, and a model that works well for one subject cannot be expected to transfer reliably to another without additional adaptation. Second, exposing the model to multiple subjects and learning the representation jointly with the classification task can improve cross-subject performance, while a small amount of subject-specific calibration provides an additional and practical improvement.

At the same time, the results should be viewed in the context of the limitations of this study, particularly the small number of subjects, the use of only the labelled `T` sessions, and the restricted EEGNet training budget. These limitations leave several important questions open, especially regarding how much the results could improve with longer training, more data, and explicit domain-adaptation or transfer-learning methods. Nevertheless, the experiments provide a clear picture of the main challenge: for MI-EEG, achieving good performance within a subject is considerably easier than building a model that remains reliable when the subject changes.

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A Additional Figures

This appendix contains supplementary diagnostic figures referenced in the main text. These figures provide additional information about the preprocessing and modelling steps but are not essential to the main results.

A.1 Experiment 1 – Supplementary Figures

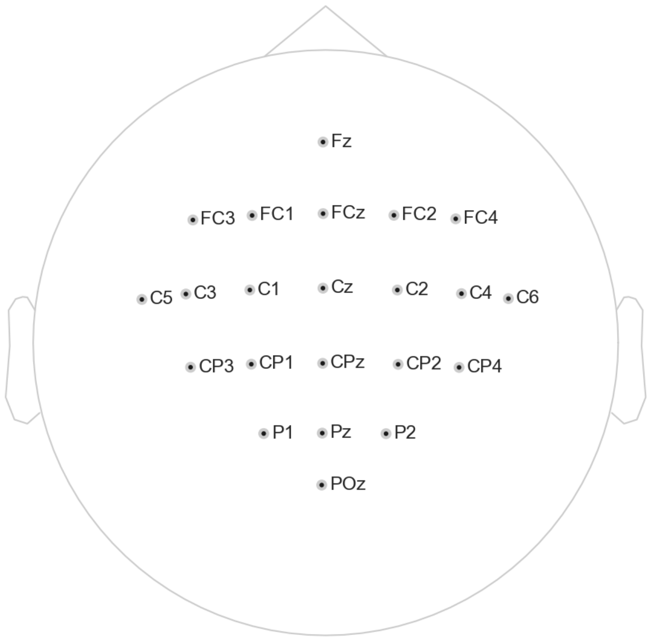

Figure 8: 10–20 electrode layout used for Subject A01T recordings, showing the 22 EEG and 3 EOG sensor positions after channel renaming.

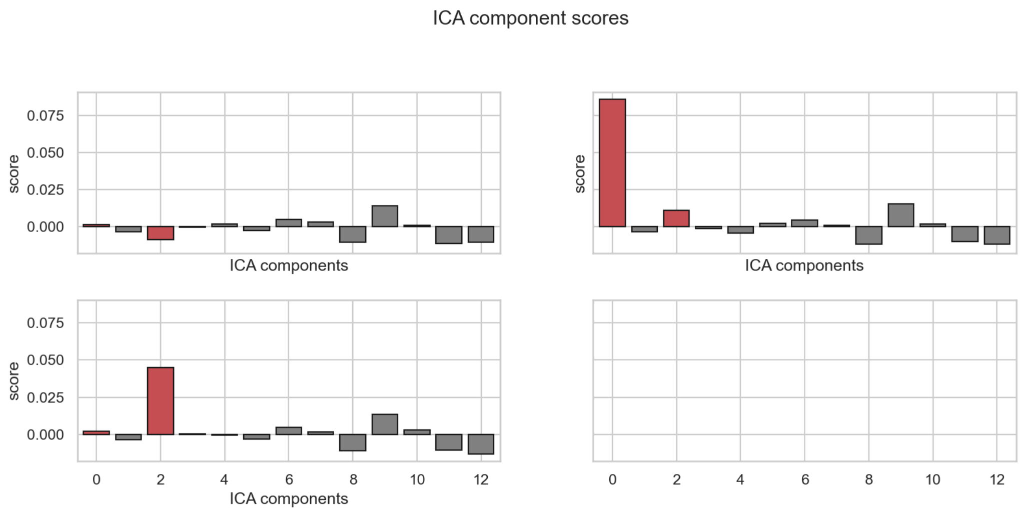

Figure 9: ICA component–EOG correlation scores used to automatically flag ocular-artifact components for removal (threshold = 2.5), Subject A01T. Flagged components are excluded from the reconstructed signal prior to band-pass filtering.

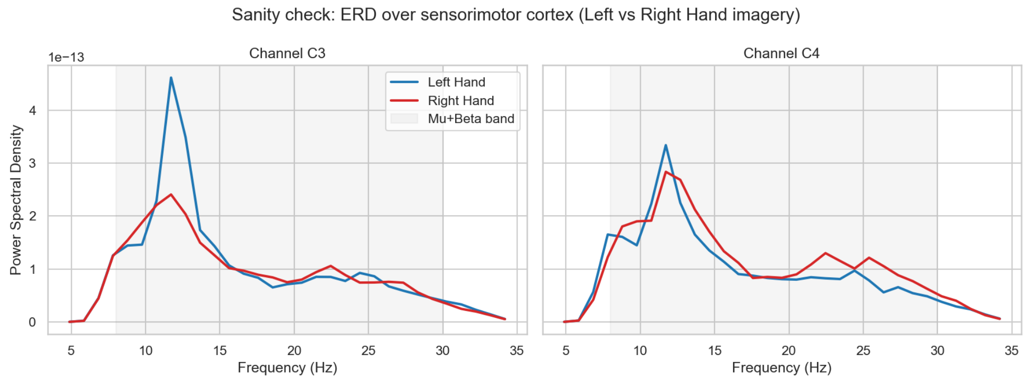

Figure 10: Sanity check for event-related desynchronization (ERD): mean power spectral density at channels C3 and C4 for left- vs. right-hand imagery, Subject A01T. The expected contralateral ERD pattern is visible within the shaded mu+beta band.

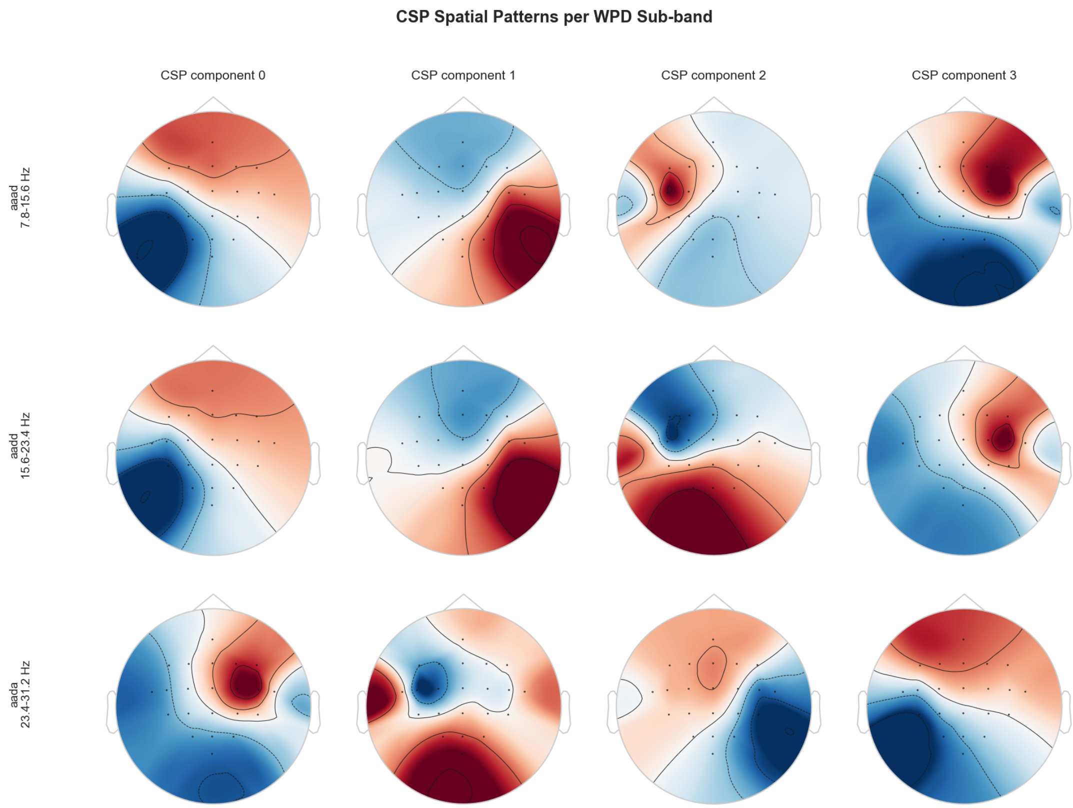

Figure 11: CSP spatial-filter topographies for each of the three relevant WPD sub-bands (rows) and the top four spatial components (columns), Subject A01T.

Figure 12: Training and validation loss/accuracy curves for the subject-dependent MLP (Subject A01T only). Early stopping halted training at epoch 65; the lowest-validation-loss weights were restored.

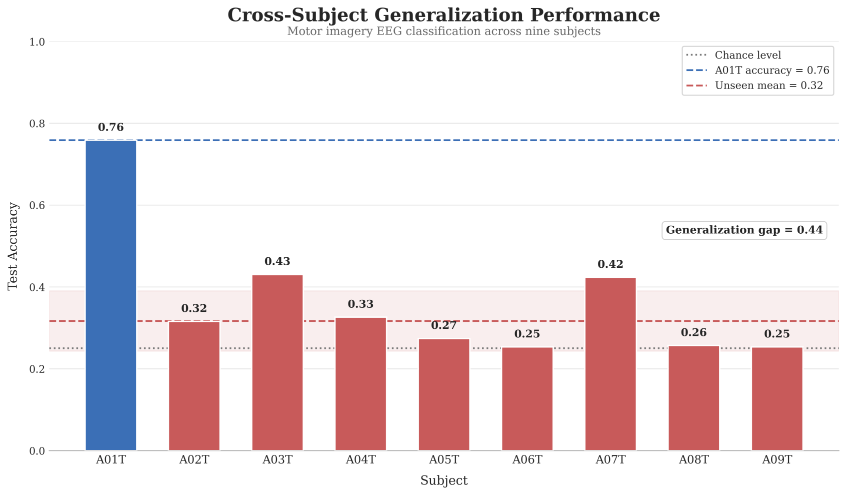

Figure 13: Overall cross-subject generalization summary: per-subject accuracy with reference lines for chance level, in-sample (A01T) accuracy, and mean unseen-subject accuracy; the shaded band indicates ±1 SD across unseen subjects.

A.2 Experiment 2 – Supplementary Figures

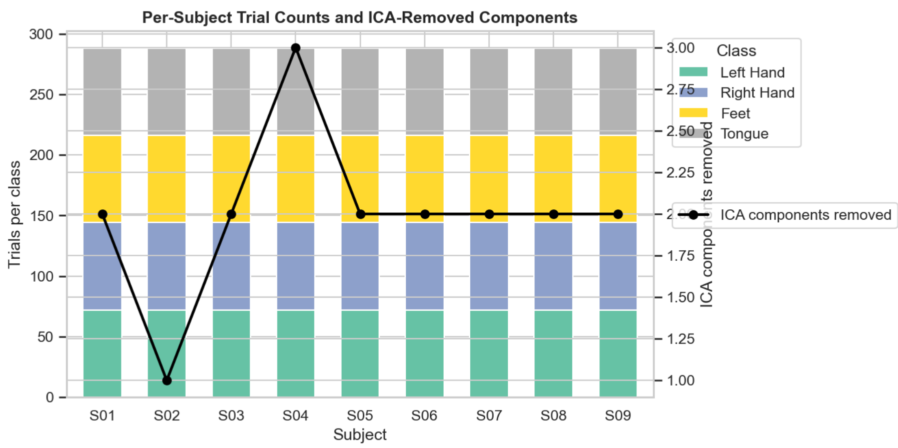

Figure 14: Per-subject trial counts by class (stacked bars) and number of ICA components removed as ocular artifacts (line), across all nine subjects.

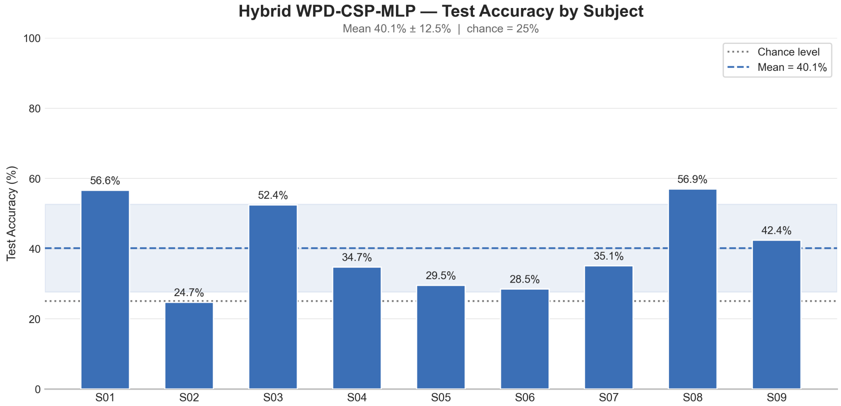

Figure 15: Simplified per-subject mean-accuracy summary for the LOSO Hybrid WPD-CSP-MLP classifier, expressed as a percentage, without the baseline or per-class breakdown shown in Figure 3.

Table 11: Experiment 2 – full per-class F1-score breakdown across all nine LOSO folds.

Test Subject	F1 (Left Hand)	F1 (Right Hand)	F1 (Feet)	F1 (Tongue)
S01	0.634	0.680	0.581	0.150
S02	0.000	0.218	0.382	0.027
S03	0.576	0.442	0.465	0.532
S04	0.286	0.419	0.026	0.443
S05	0.345	0.388	0.182	0.000
S06	0.053	0.000	0.225	0.453
S07	0.438	0.454	0.301	0.000
S08	0.528	0.639	0.548	0.556
S09	0.482	0.396	0.140	0.500

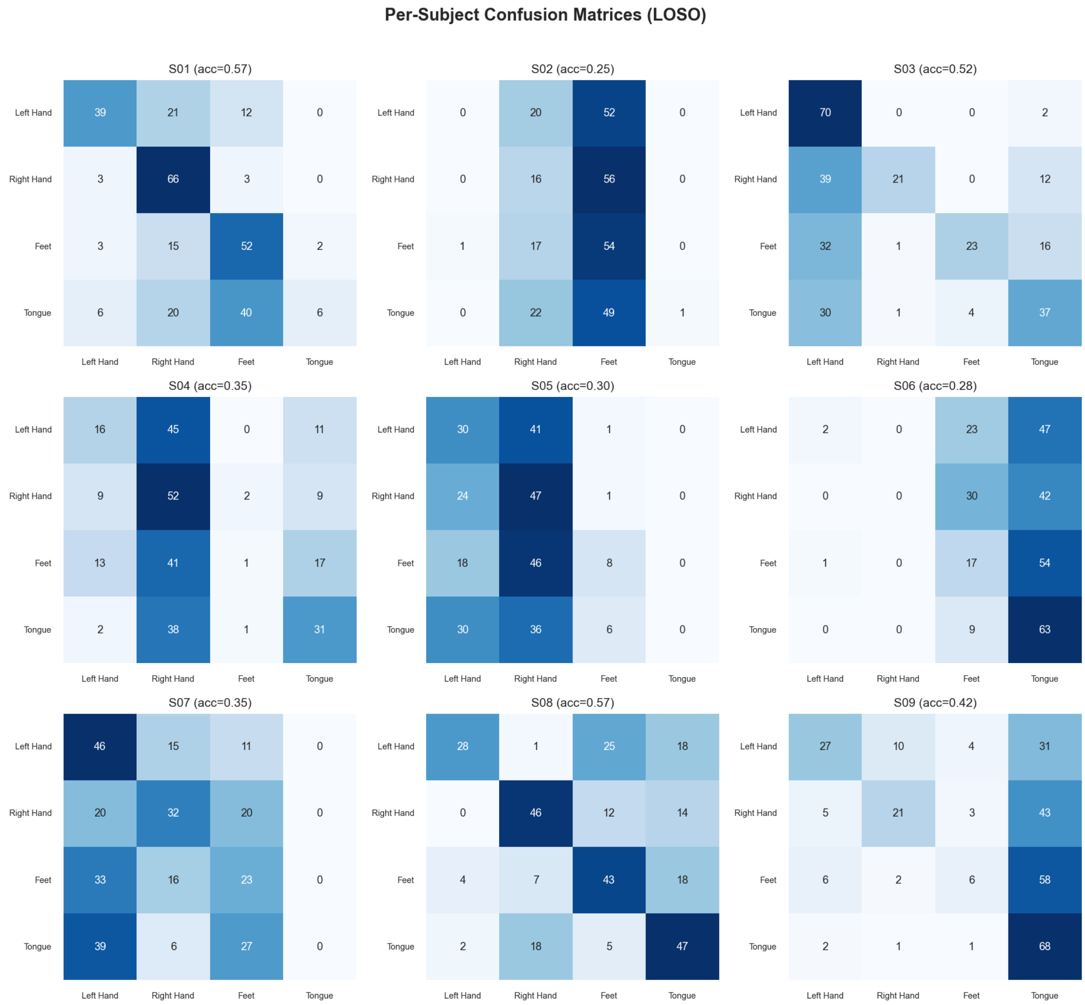

Figure 16: Per-subject confusion matrices for all nine LOSO folds (Hybrid WPD-CSP-MLP).

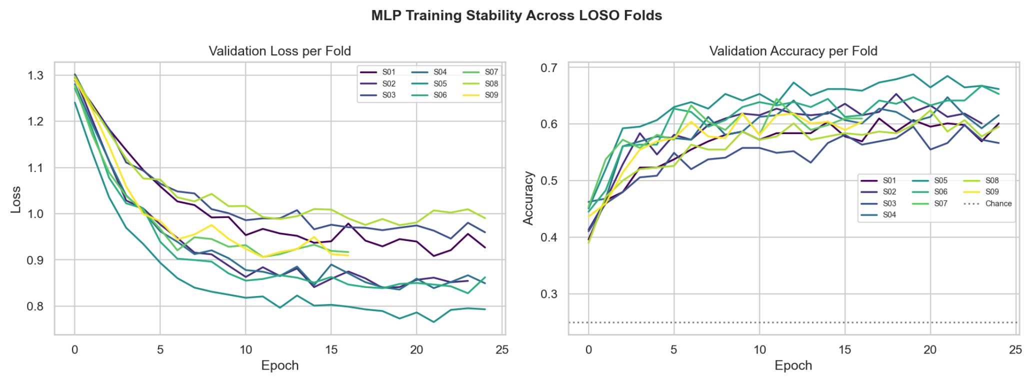

Figure 17: Validation loss and accuracy trajectories for the MLP across all nine LOSO folds, illustrating training stability across subjects.

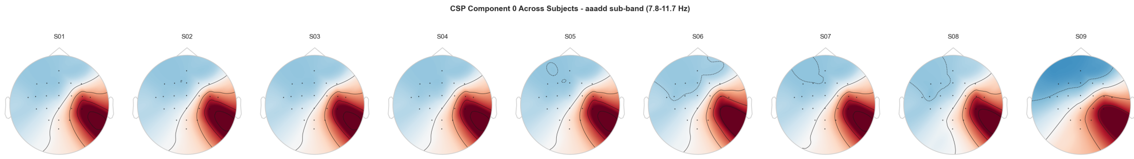

Figure 18: CSP component 0 topography (first relevant sub-band) fitted independently for each subject, illustrating inter-subject variability in the optimal spatial filter – a visual counterpart to the domain-shift result of Figure 5.

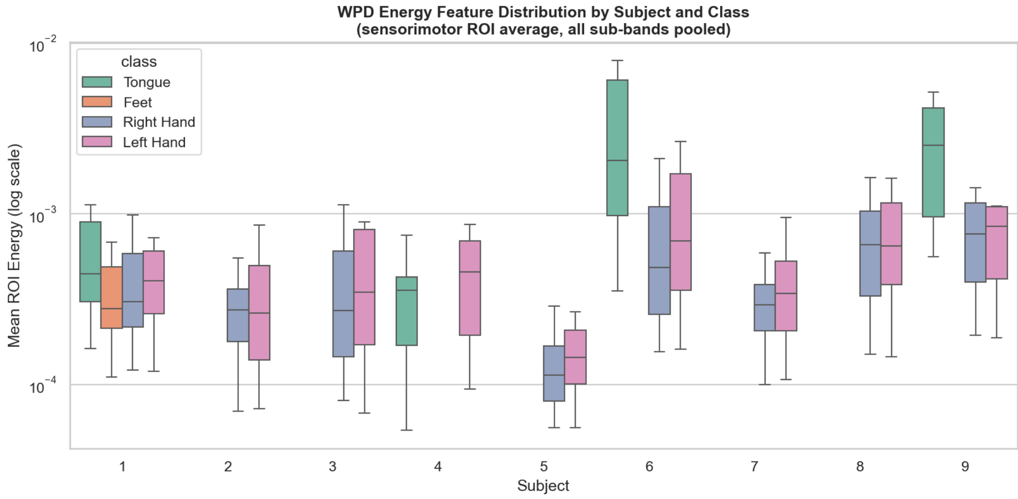

Figure 19: WPD energy feature distribution (sensorimotor ROI average, pooled sub-bands) by subject and class, illustrating heterogeneous energy scales across subjects.

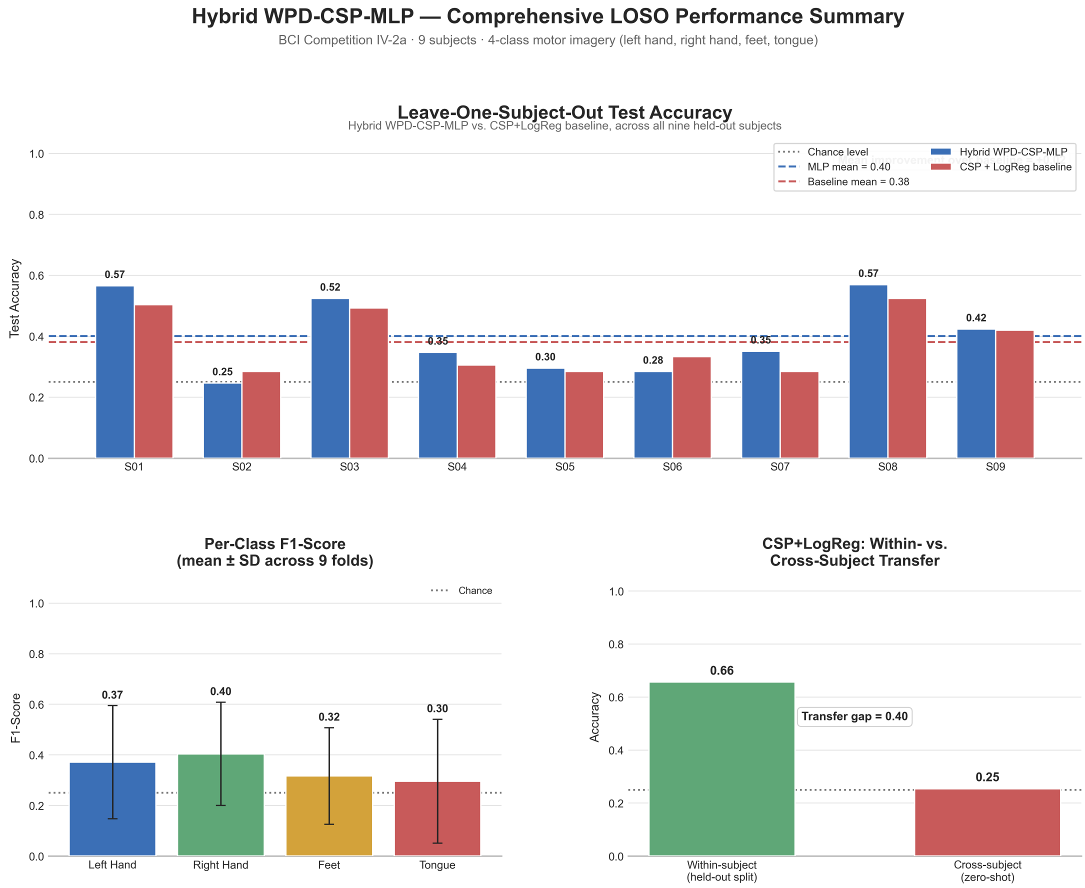

Figure 20: Consolidated LOSO results dashboard summarizing per-subject Hybrid-MLP vs. baseline accuracy alongside auxiliary summary statistics.